

Validation criteria that exclude their own mechanism: six failure modes of voxel damage operators applied to Ni-YSZ tomograms

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Abstract

Applying a voxel-scale damage operator to a segmented tomogram is a standard way to simulate microstructural degradation in solid oxide cell electrodes. We asked whether any operator of this class reproduces a sharp measured signature in three FIB-SEM reconstructed Ni-YSZ anodes spanning a coarseness series: the finest anode retains nickel percolation worst (0.680 against 0.855 and 0.947) yet retains triple-phase boundary best. None does. The reasons are methodological, and they are the useful part of the result.

The most general is an impossibility. Requiring an operator to reduce specific surface area monotonically is the natural way to validate a coarsening rule, and it excludes Rayleigh-type neck break-up, the mechanism such rules exist to represent. On a 6-connected lattice a single-voxel swap changes exposed area by exactly $2[nb(a) - nb(b)]$, so the criterion is a statement about neighbour counts. A curvature-ranked operator thins a model neck from 63 voxels to zero but violates the criterion at its first step; a greedy area-decreasing operator never violates it and accepts no move at all, because a sphere-neck-sphere body is already a local minimum under single-voxel displacement. Only finite-temperature acceptance bridges the two, and a monotonicity gate forbids exactly that.

Five further failure modes are measured. Retaining the largest connected component drives the spanning fraction to unity identically, so the operator rewrites the denominator of the metric used to judge it. Voxel-scale erosion multiplies triple-phase boundary density seven- to fifteen-fold before destroying it. Qualification gates demanding perfect pristine connectivity are unrepresentative of real electrodes, which carry 1.2% to 11.2% disconnected nickel. A jittered regular lattice has a minimum cut of exactly one full cross-section with zero seed-to-seed variance, whereas real networks fail at 1% to 2% of throats through small non-planar cuts. Curvature-ranked moves do not deliver the area reduction they are assumed to deliver. Each was invisible on synthetic test structures and each would silently corrupt a published study of this kind.

Keywords: solid oxide cell, Ni-YSZ anode, FIB-SEM tomography, percolation, triple-phase boundary, microstructural degradation

1. Introduction

Nickel-yttria-stabilised-zirconia cermet anodes lose performance in service through microstructural change. Nickel coarsens and can lose the percolating network that carries electronic current [1]; the ceramic scaffold fractures under redox cycling [2]; and the triple-phase boundary (TPB), the one-dimensional locus where nickel, zirconia and pore meet and where the electrochemical reaction proceeds, is consumed [3]. All three are geometric processes acting on a microstructure that FIB-SEM and X-ray nano-tomography now resolve routinely, which makes a particular computational strategy very attractive: take a segmented tomogram, apply a damage operator to the label field, and track how connectivity and TPB evolve.

The strategy is attractive because it is cheap and because it acts on the same data structure the experimentalist al-

ready has. It is also, as far as we can tell, less examined than the continuum alternative. Phase-field treatments of Ni coarsening in this system have been developed and progressively validated over more than a decade, from the early three-phase simulations on experimentally acquired anode functional layers [4], through explicit validation studies against ex-situ imaging of the same specimen [5], to quantitative simulations initialised on ptychographic nano-tomograms with measured thermophysical parameters [6] and multiphase-field models coupled to transmission-line performance estimates on FIB-SEM reconstructions [7]. Discrete methods have been applied on the fabrication side, where a kinetic Monte Carlo model of NiO-YSZ sintering was calibrated and validated against measured microstructure [8]. What is comparatively thin is a systematic account of what happens when a discrete, voxel-level operator is used for the degradation problem specifically, and of how the conventions that surround such an operator—the connectivity metric, the qualification gate on the test structure, the validity criterion on the operator itself—interact with

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each other.

This paper is that account, and it is negative in its first-order result. We set out to identify which damage mechanism reproduces a measured degradation signature in three Ni-YSZ anodes, and we did not find one. Two principal geometry-based operator classes were tested to a conclusion and neither reproduced the ordering; two further routes closed before a mechanism test was reached. Along the way six specific ways in which this class of simulation silently misleads were identified and measured. We think the second finding is the more useful one, and the paper is organised around it, but the two are not separable: the failure modes were found by running the programme that failed, and they are reported in the order in which the work forced us to confront them.

The experimental target is unusually sharp and runs against intuition. Across the anode series of Pecho and co-workers [9, 10] the finest microstructure retains nickel percolation *worst* while retaining TPB *best* (Fig. 1, Table 1). A coarsening argument predicts the first half: by Herring’s scaling laws the time to produce a geometrically similar change scales as a power of the particle size [11], so the finest structure should coarsen fastest and lose its network first. It does not obviously predict the second half, in which the same anode preserves four fifths of its reaction sites.

Two cautions travel with that target throughout this paper, and both follow from how the data were acquired rather than from how they were analysed. Medium versus coarse is not resolved on the nickel side: the ordering flips depending on whether percolation is defined by two-face spanning or by reachability from one face. And the published TPB series is not monotone, since coarse (0.608) exceeds medium (0.586). Only the two-level statements—*fine retains nickel percolation worst* and *fine retains TPB best*—are defensible, and those are the only claims we test against.

2. Materials and methods

2.1. Tomograms

Six segmented FIB-SEM stacks were used: three anodes, designated fine, medium and coarse by mean nickel particle diameter, in pristine and post-redox states. Stacks are 0.48 to 1.11 gigavoxels at pitches of 19.53 nm to 29.14 nm [9, 10]. Phase labelling was verified against the published volume fractions, with a worst deviation of 0.20 % across eighteen values.

The pristine and degraded stacks are different specimens (Rx36/Rx37/Rx38 against Rx41-1/Rx41-2/Rx41-3), not one volume imaged twice. Every retention magnitude is therefore confounded with specimen-to-specimen variation, and no paired before-and-after quantity is computed anywhere in this work. This is the principal reason all comparisons here are ordinal.

2.2. Percolation and connectivity

Phase connectivity uses 6-connectivity with free, non-periodic boundaries. A cluster percolates if a single connected component reaches both extreme slices of the transport axis; P_{span} is the fraction of phase voxels in such a cluster, and P_{reach} the fraction reachable from one face. The implementation recovers the simple-cubic site-percolation threshold as 0.31218 after finite-size extrapolation, against a reference value of 0.3116077 [12].

2.3. Network extraction

Particle networks were extracted by watershed partitioning of the Euclidean distance transform of the solid phase [13]. Nodes are nickel chambers; edges are throats carrying a measured inscribed diameter and cross-sectional area. A defect in the library’s default parallel dispatch was found and corrected during this work; its quantified effect on the earliest extractions is smaller than the between-region spread and changes no conclusion reported here.

2.4. TPB estimation

TPB density is the total length of edges shared by three differently labelled voxels, per unit volume. Gathering the four voxels around an edge with a circular shift wraps at the domain faces and manufactures triple lines; on an analytic single-triple-line test case the wrapping implementation over-counted by exactly a factor of four. The corrected estimator restricts the gather to interior edges. Estimates of TPB length from voxelised data carry a known systematic error whose size depends on the calculation rule and on the resolution [14, 15]; we therefore report TPB as a ratio to the pristine value of the same region wherever a conclusion depends on it, and quote absolute densities only for comparison against published values.

2.5. Damage operators

Two operator families were run to a conclusion. The first is stochastic surface erosion: at each round, every nickel voxel on the phase boundary is removed independently with probability $p_{\text{erode}} = 0.35$, uniformly over surface sites with no curvature weighting, after which the largest connected component is retained. A variant carrying an oxidative dilation step was tested and rejected on validity grounds (Section 3.3). The second is volume-conserving coarsening, in which a nickel voxel is removed from one site and added at another in the same round, so that the phase volume is exactly preserved. Two move-selection rules were implemented, both pre-registered: ranking candidate sites by a curvature proxy, and greedy acceptance of a move only when it does not increase exposed area.

Three further operators were run on a synthetic platform whose nickel phase is a jittered regular lattice of spheres joined by explicit necks and whose YSZ phase is a particulate generator with an adjustable sintering yield. Initial yields were set from simple-cubic bond-percolation priors [16] and then recalibrated once, before any damage,

against the measured connectivity of the generator’s own face-centred-cubic contact graph. The three operators were: widening the lower tail of the neck-size distribution at conserved nickel loading, severing every throat in the lower quartile of the size distribution, and progressive fracture of sintered YSZ contacts. A cumulative-strain YSZ fracture operator was run on real tomograms. Definitions, frozen parameters and unit tests are in the repository.

2.6. Outcome metric

Following the discovery described in Section 3.3 we adopted

$$R_{\text{Ni}} = \frac{\text{spanning-cluster nickel voxels}}{\text{pristine nickel voxels}}, \quad (1)$$

which is invariant to the deletion of isolated material, monotone non-increasing under any removal operator, and directly comparable to measured retention. Its mandatory sanity check—that R_{Ni} evaluated on the undamaged structure equals the pristine spanning fraction—holds to 0.00×10^0 in all three anodes.

2.7. Statistics and pre-registration

There are three anodes. No p -values are reported anywhere in this work, and none should be computed from three points. Where correlations appear they are Spearman rank coefficients with leave-one-out sign checks against a threshold of $|\rho| \geq 0.6$ fixed in advance. Region-to-region and seed-to-seed spread is reported as a range rather than as an interval with a coverage claim.

Each phase of the work was pre-registered in a version-controlled document committed before the run it governed, specifying the operator definition, frozen parameters, validity gates, decision rules and failure branches. Where a pre-registered rule later proved ill-chosen it was honoured and the defect reported, rather than amended after the fact. Two consequences of that policy appear below as reported nulls that a re-picked parameter would have avoided.

3. Results

3.1. The target

Table 1 and Fig. 1 give the outcome. Our absolute values track the published ones closely: P_{reach} matches the published P to within 0.1 % to 7.4 % on all six stacks. The one fact that survives every definition is that the fine anode retains nickel percolation worst, and it survives because the source papers state it independently. The fine anode is simultaneously the best TPB retainer, which is what makes the target interesting: whatever mechanism destroys the nickel network in the finest structure leaves most of its reaction sites intact.

3.2. A synthetic platform, and what it concealed

Work began on a synthetic three-phase platform: nickel spheres on a jittered regular lattice joined by explicit necks, with a particulate YSZ generator whose sintering yield sets ceramic connectivity independently of grain size. The platform passed a twelve-part qualification suite on volume fractions, percolation, particle sizes, neck sizes, length-scale ordering and ceramic fragility ordering. It was, by every check we thought to apply, a good analogue.

Three mechanisms were tested on it and none produced the target ordering. Widening the lower tail of the neck-size distribution by 1.33 and 2.00 times at conserved nickel loading moved the bisection midpoint from 8.54 rounds to 8.50 and 8.50, a change of -0.04 rounds against a pre-registered interpretability threshold of 1.0; 74 of 75 bisections returned exactly 8.5. Severing every throat in the lower quartile of the size distribution—100% of candidates, at a cost of 0.6 % to 5.3 % of the nickel volume—produced no percolation failure whatever, with $P_{\text{span}} = 1.0000$ at the maximum intensity in 15 of 15 structures. Progressive fracture of sintered YSZ contacts did separate the classes, by 2.87 rounds in the predicted direction with non-overlapping seed ranges, and a matched-fragility control in which pristine ceramic fragility was equalised across classes inverted that separation to -0.27 rounds. The apparent mechanism was pristine-state loading: the class calibrated to start nearest its percolation threshold failed first, as it must. Without the control the main-arm result would have read as a confident positive.

A graph audit run to explain the first two nulls found something more general than the nulls themselves. The minimum source–sink cut of the synthetic nickel network is exactly one complete lattice cross-section—36, 25 and 16 edges for the three classes, which are 6^2 , 5^2 and 4^2 —in every one of fifteen structures, with zero seed-to-seed variance (Fig. 2). A jittered regular lattice has no locally weak plane: every cross-section is statistically identical, so the minimum cut is a plane and its size is fixed by the geometry rather than by the microstructure. Removing half of all necks in random or increasing-area order still leaves 91 % to 97 % of the nickel in the spanning cluster, while a targeted cut breaks the network at 5.8 % to 7.1 % of throats, a factor of about eight apart. The critical edges overlap the lower-quartile neck population at 0.250, 0.280 and 0.188 against a chance expectation of 0.25, which is why severing the narrow necks did nothing: in this architecture the narrow necks are not load-bearing.

The same audit measured the real networks for comparison. Real nickel networks are disordered in a way the lattice is not: connected-pair-distance coefficients of variation of 0.38 to 0.45, coordination standard deviations of 1.8 to 2.9 against a lattice whose interior coordination is fixed, and minimum cuts of a handful of edges rather than a plane. This is the first of the failure modes, and it is a statement about what a synthetic platform can be asked to do. A regular lattice cannot fail the way a real network

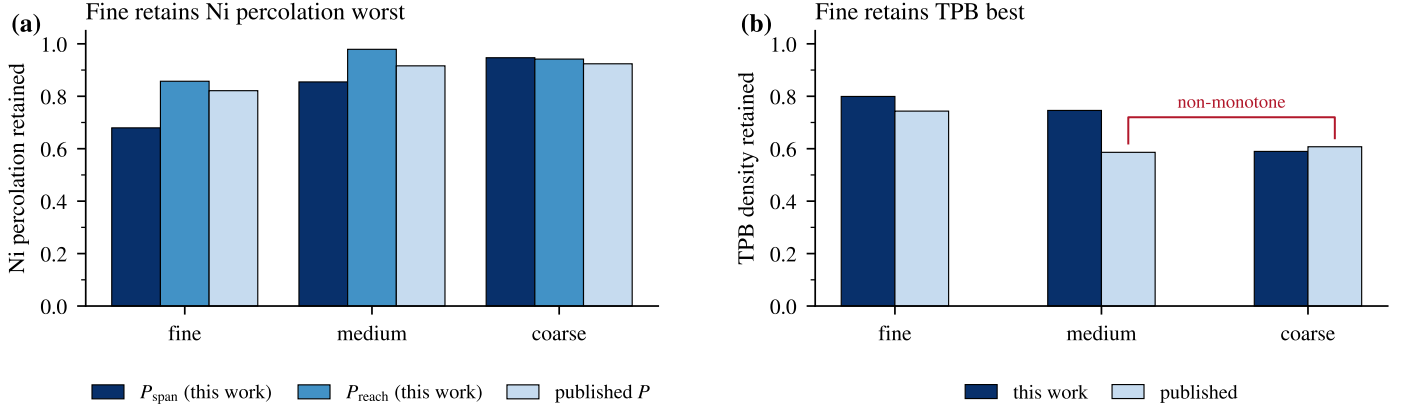


Figure 1: The measured signature the operators were built to reproduce. (a) Retained nickel percolation under three definitions. Fine is worst on all three; medium and coarse exchange places between P_{span} and P_{reach} , so that comparison is treated as unresolved. (b) Retained TPB density. Fine is best in both series, but the published series is not monotone, so only the two-level statement is used. Data: phase6_comparison_table.csv.

Table 1: Measured retention, degraded divided by pristine, on full stacks with no sub-sampling. Published values are from [9, 10]. Pristine and degraded are different specimens.

anode	Ni percolation retained			TPB retained	
	P_{span}	P_{reach}	published P	this work	published
fine	0.680	0.857	0.821	0.799	0.743
medium	0.855	0.979	0.916	0.746	0.586
coarse	0.947	0.942	0.924	0.590	0.608

fails, and no local stochastic operator can make it do so. Bond dilution does not help, because a diluted regular lattice remains statistically homogeneous at large scale and the cut plane merely thins; the property the lattice lacks is positional disorder, which cannot be created by deleting bonds from a periodic grid.

The second failure mode surfaced only when the operators were moved to real data. Measured pristine spanning fractions per region are 0.9821/0.9754/0.9877 (fine), 0.9713/0.9446/0.9554 (medium) and 0.8878/0.9528/0.9157 (coarse). Between 1.2 % and 11.2 % of the nickel in a real electrode is already disconnected before service, and the disconnected fraction is coarseness-ordered. The published percolation values carry the same message less sharply, at 0.959 to 0.985 rather than unity. Our synthetic qualification gate required pristine P_{span} to equal 1.0000 exactly, and every structure passed it. That gate is unrepresentative of every real electrode measured here, and it is what hid the next failure mode for the entire synthetic phase of the work.

3.3. The outcome metric rewrites its own denominator

The spanning fraction divides spanning-cluster voxels by total phase voxels. Damage operators conventionally retain only the largest connected component, on the physical grounds that isolated nickel is electrically dead. The two conventions are individually reasonable and jointly circular. Pruning deletes exactly the non-spanning voxels, so after

pruning the surviving component *is* the spanning one and P_{span} equals unity identically. The operator rewrites the denominator of the metric used to judge it. This is the third failure mode and the one that cost us the most time.

On real regions, pristine P_{span} of 0.9821, 0.9713 and 0.8878 became exactly 1.0000 at the first damage step, in two independently implemented operators (Fig. 3a). Our first diagnosis attributed this to an oxidative dilation step, on the reasoning that dilating nickel into pore would bridge the disconnected fraction. That diagnosis was wrong. Removing the dilation step changed almost nothing—the fine anode’s TPB inflation moved from 15.24 to 14.80 and P_{span} remained exactly 1.0000—because the cause is the pruning step, not the dilation.

This is a metric-operator incompatibility rather than an operator defect, and the fix belongs to the metric. Scoring on spanning-cluster voxels as a fraction of *pristine* nickel gives a quantity that does not move when isolated material is deleted, falls monotonically under erosion (Fig. 3b), and is directly comparable to the measured retention values. On the synthetic platform the pathology was invisible for a precise reason: the qualification gate had forced pristine P_{span} to unity, leaving the pruning step nothing to remove.

3.4. Voxel-scale erosion manufactures TPB before destroying it

The fourth failure mode is in the same runs. TPB density is multiplied by 14.8, 12.3 and 7.4 at the first damage round

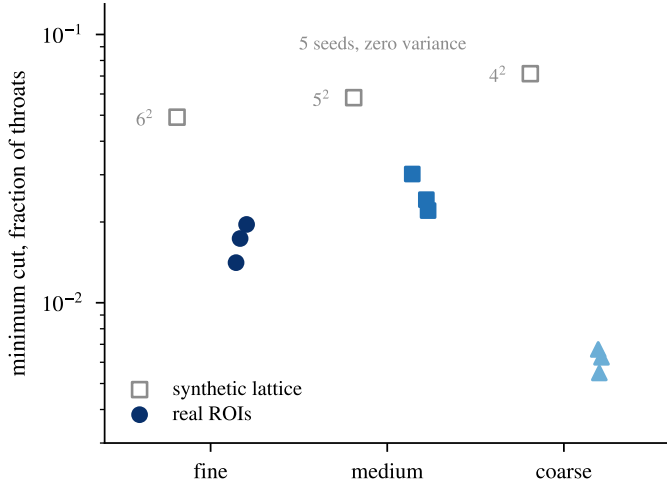


Figure 2: Minimum source-sink cut as a fraction of all throats, logarithmic axis. The synthetic lattice cuts at exactly one full cross-section, at the same fraction in all five structure seeds of each class. Real regions of interest cut at 1% to 2% of throats with genuine region-to-region spread. Data: `audit_ni_vulnerability.csv`, `cireal_results.csv`.

before it collapses to a few percent of pristine by the eighth (Fig. 4). The mechanism is discretisation: removing single voxels from a surface roughens it at the voxel scale, and each newly exposed nickel/pore facet that happens to touch zirconia adds triple line. The volume-conserving operator described below showed the same signature more mildly, doubling TPB while increasing nickel surface area by 3.3%.

That voxelised TPB estimates carry a resolution-dependent systematic bias is established [14, 15]. What is different here is that the bias is not a fixed offset but a large, non-monotone excursion driven by the operator itself, so it cannot be cancelled by taking ratios to the pristine state or by any calibration performed before damage. An operator whose TPB retention exceeds unity is roughening rather than eroding, and this is worth stating as a routine check: it costs one line and it disqualified three operators here. On our synthetic platform the same excursion was inconspicuous because the platform’s absolute TPB was already about eightfold too high, an artifact of synthetic phase placement rather than of the estimator. On real data the corrected estimator returns 4.48 to 4.66, 1.87 to 2.37 and $1.47 \mu\text{m}^{-2}$ to $1.55 \mu\text{m}^{-2}$ for the three anodes against published values of 3.62, 2.11 and 1.47, a factor of 1.0 to 1.3.

3.5. Surface erosion reverses the measured ordering

With a pruning-invariant metric and an operator that passes its remaining validity checks, the mechanism test can finally be run. It fails, and it fails in the wrong direction.

Bisecting the damage intensity at which R_{Ni} falls below 0.50 gives class means of 9.72 (fine), 9.06 (medium) and 9.39 (coarse) rounds; at a threshold of 0.10 the values are 9.72, 9.28 and 9.39. The pre-registered criterion required fine to be lower than both others by at least one round.

Fine is higher at both thresholds, and every separation is also below one round, so the result is simultaneously wrong in sign and unresolved in magnitude (Fig. 5a,b). Against the measurement this is a complete reversal: real retention runs $\text{fine} < \text{medium} < \text{coarse}$ from worst to best, while the simulation gives $\text{medium} < \text{coarse} < \text{fine}$.

Two candidate explanations were measured directly and neither survives. A kinetic hypothesis predicts that the anode with the largest specific nickel surface area erodes fastest; the raw Spearman coefficient is +0.018 at the 0.50 threshold and +0.089 at 0.10 (Fig. 5c). A topological hypothesis predicts that the anode with the smallest minimum cut fails first; the coefficients are +0.028 and +0.213. Nothing reaches the pre-registered $|\rho| \geq 0.6$, partial correlations controlling for pristine P_{span} change no sign that matters, and leave-one-out signs are inconsistent for three of the four combinations.

The minimum-cut measurement produces a result worth separating from the null. With the coarse regions properly sized at $12 \mu\text{m}$ (258 to 299 graph nodes, against 48 to 72 in an earlier under-sized pass), the coarse anode is by a wide margin the most topologically fragile of the three: minimum cut fractions are 0.0141 to 0.0196 (fine), 0.0221 to 0.0302 (medium) and 0.0055 to 0.0067 (coarse). Coarse is also the best retainer in service, at 0.947. Pristine topological fragility runs three to five times against the outcome it is supposed to predict. Taken with the absent surface area correlation, the sharpest statement this dataset supports about the measured ordering is a statement about what does not control it: neither the pristine topology nor the specific surface area does.

Why the erosion operator behaves as it does is visible in the synthetic runs, where the volume loss at the transition was recorded. Fine loses 74.4% of its nickel before the network breaks, medium 70.5% and coarse 62.8%. Fine does erode faster, and its 1.26-fold specific-surface-area advantage is real, but its network tolerates proportionally more volume loss before it stops spanning. High specific surface area and high connectivity redundancy are both consequences of fineness and they oppose each other in this outcome variable, cancelling almost exactly.

3.6. The area barrier

Surface erosion removes volume. Real coarsening does not: it conserves nickel and reduces surface area. A removal-only operator also cannot reproduce the measured change in nickel volume fraction, which is -0.1006 , -0.0165 and $+0.0146$ across the three anodes and therefore *increases* in the coarse case. So the natural next operator is volume-conserving, and it is the one that produces the most general result in this paper.

For a single-voxel swap on a 6-connected lattice the change in exposed nickel area is exact rather than approximate. Removing a surface voxel a changes the count of exposed nickel faces by $2 \text{nb}(a) - 6$, where nb counts nickel neighbours, and adding a voxel at a pore site b changes it

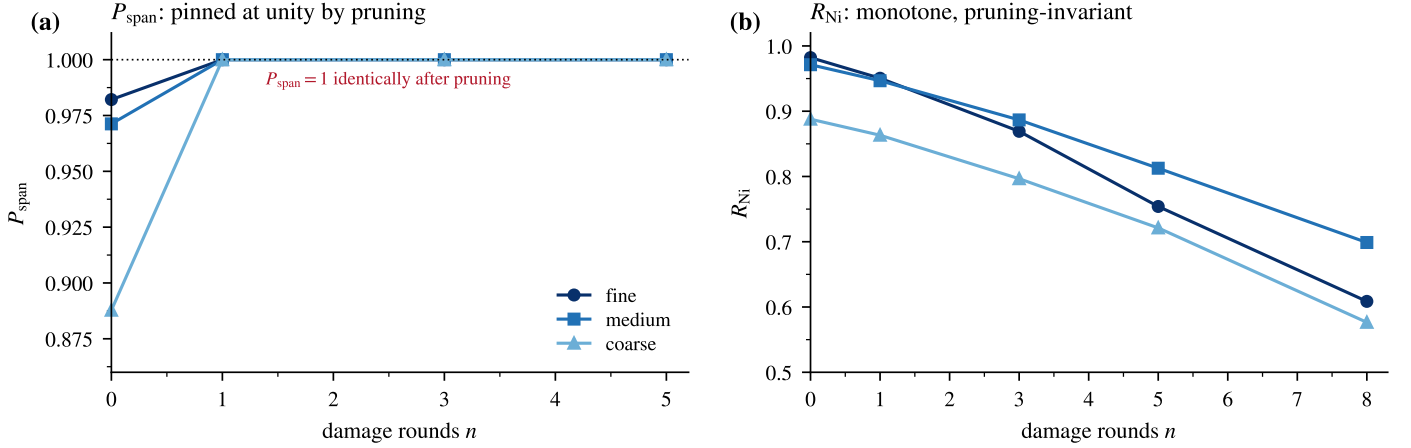


Figure 3: (a) Spanning fraction under surface erosion on real regions of interest. Every anode reaches exactly unity at the first damage round and stays there, because largest-component pruning deletes the non-spanning voxels that form the denominator. (b) The pruning-invariant metric R_{Ni} on the same runs declines monotonically from a pristine value that is not forced to unity. Data: `cireal_o6_validity.csv`, `cireal_rni_gate.csv`.

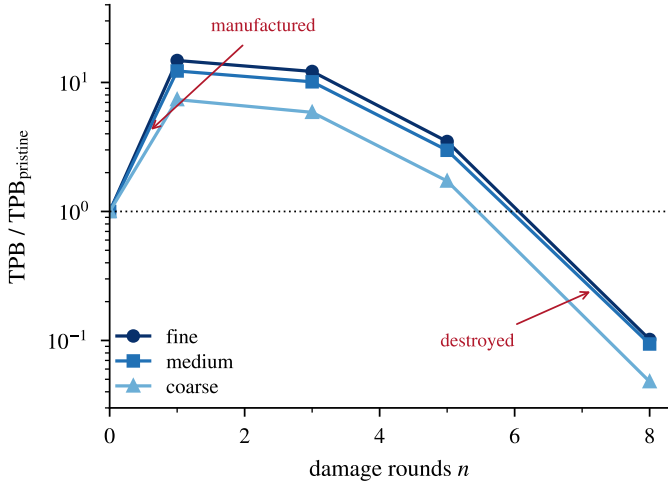


Figure 4: TPB density relative to pristine under surface erosion on real regions of interest. Stochastic single-voxel removal pits the nickel surface, and every new nickel/pore facet touching zirconia creates triple line. Any TPB conclusion drawn before the peak reports discretisation, not the electrode. Data: `cireal_rni_gate.csv`.

by $6 - 2\text{nb}(b)$. Hence

$$\Delta A = 2[\text{nb}(a) - \text{nb}(b)], \quad (2)$$

so that $\Delta A \leq 0$ if and only if $\text{nb}(a) \leq \text{nb}(b)$. Material must leave a site with few nickel neighbours, which is convex and of high curvature, and arrive at one with many, which is concave (Fig. 6a). The expected direction of matter flow is recovered without approximation.

A coarsening operator ought to reduce specific surface area, so requiring it to do so monotonically is the natural validity gate, and we pre-registered exactly that gate. Two move-selection rules were implemented against it and they bracket the space of single-swap algorithms between them.

The first ranks candidate moves by a curvature proxy and applies them without testing the area change. With

a 6-connectivity stencil it does nothing useful, leaving the model neck at 63 voxels; with a 26-connectivity stencil the neck thins monotonically, 63 to 57 to 15 to 0, at exactly conserved volume (Fig. 6b). This is genuine Rayleigh-type behaviour, and it fails the gate, because at the first round the surface area *rises*, to 0.45195 against a pristine 0.45052, under both stencils (Fig. 6c). Curvature rank is a proxy for the area change and not the area change itself: on a discrete lattice the added voxel exposes new faces at its own free sides, and at $n = 1$ that transient exceeds the area removed. By $n = 3$ the added material has coalesced into the neck and the net area falls below pristine. This is a fifth failure mode in its own right, and it is the one that Eq. (2) was derived to repair.

The second rule uses Eq. (2) directly, accepting a swap only when $\Delta A \leq 0$. It cannot violate the gate by construction, and it accepts zero moves. Not a small number—zero, at every intensity tested, so that the structure is returned bit-identical and the gate fails on the requirement of a *strict* decrease. The reason is not an implementation fault. In a sphere–cylinder–sphere body the least convex surface voxel still has more nickel neighbours than the most concave available pore site, $\text{nb}_{a,\text{min}} > \text{nb}_{b,\text{max}}$, so every candidate pair is rejected at the first comparison. The body is already a local minimum with respect to single-voxel moves. Spheres minimise area at fixed volume and a straight cylinder is stable against any individual voxel displacement.

That is the whole of the argument, and its consequence is a structural one. Rayleigh-type break-up of a solid cylinder by surface diffusion is a collective, long-wavelength instability [17]: a cylinder is unstable to perturbations of wavelength greater than its circumference, but the instability only lowers energy once a correlated set of voxels has moved together, and any single-voxel perturbation raises the area first. One operator crosses that barrier without pricing it and violates the criterion; the other prices it and refuses to cross, and does nothing. The only bridge

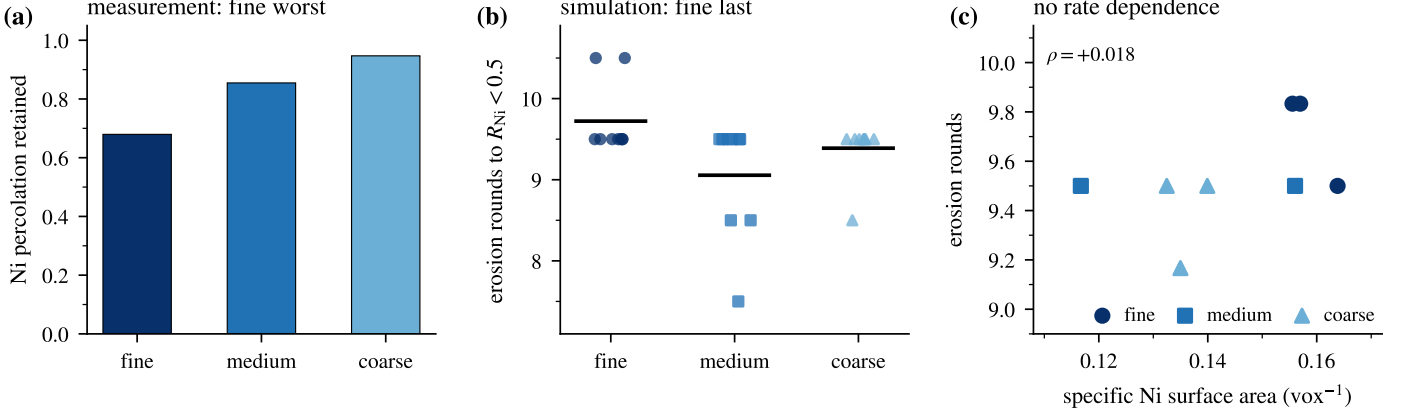


Figure 5: (a) Measured nickel percolation retention: fine is worst. (b) Erosion rounds to $R_{Ni} < 0.5$ on the same real microstructures, three regions and three damage seeds per anode; bars are class means, points are individual bisections. Fine is the *last* anode to fail. (c) Transition against pristine specific nickel surface area, one point per region. Data: `c1real_results.csv`, `phase6_comparison_table.csv`.

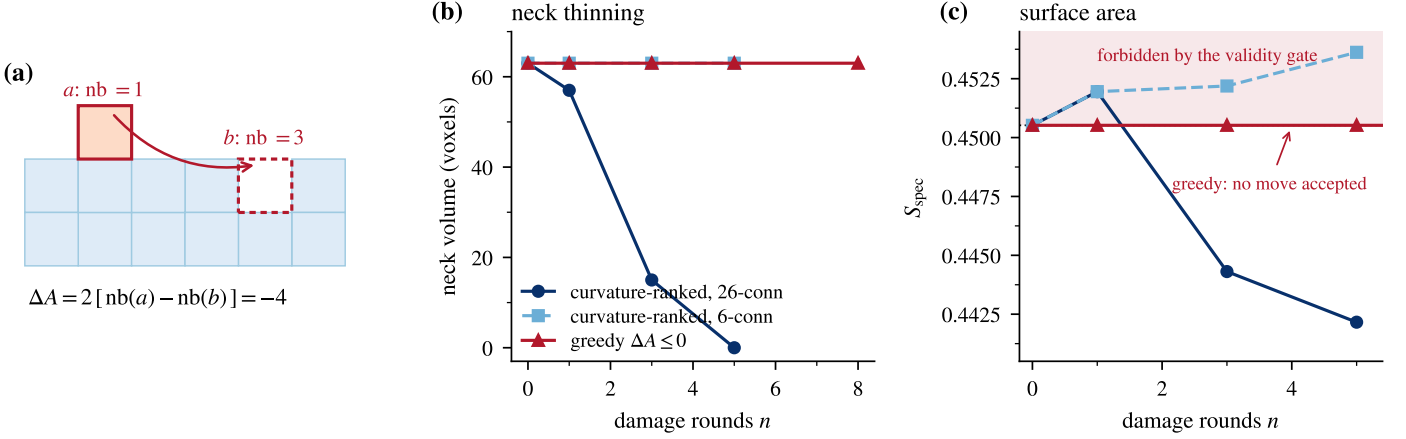


Figure 6: Two operators bracket an impossibility on a sphere-neck-sphere test body (neck 63 voxels, pristine $S_{\text{spec}} = 0.45052$). (a) The exact area change of one swap on a 6-connected lattice, Eq. (2): moving a voxel from a protrusion to a notch lowers exposed area by four faces. (b) Neck volume. The curvature-ranked operator with a 26-connectivity stencil thins the neck to zero; with 6-connectivity it does nothing; the greedy operator does nothing. (c) Surface area. The curvature-ranked operator enters the region forbidden by its own validity criterion at $n = 1$, at 0.45195 against a pristine 0.45052, before falling below pristine at $n = 3$. The greedy operator never leaves the pristine line because it accepts no move at all. Data: `o5v2_area_barrier.csv`.

is finite-temperature acceptance of area-increasing moves, which is exactly what a monotonicity gate forbids. Any voxel-based coarsening study that both enforces monotone surface-area reduction and claims to reproduce Rayleigh-type neck break-up is therefore either not producing that break-up or is violating its own validity criterion. The two cannot hold together.

What concealed this on the synthetic platform is simple: the operator was never asked to thin a neck against an explicit area budget, because the structures were fully connected and the validation was about connectivity. We stopped at this point, as the pre-registration required, and the sixth failure mode is the stopping condition itself. The agglomeration hypothesis is not falsified by any of this. It is untested, because the operator class available to us cannot represent it under a gate we still believe is the right gate for that operator class.

3.7. Two hypotheses these data cannot pose

Two further hypotheses were pre-registered with explicit untestability branches, and both branches fired. We report them because a pre-registered untestable outcome is information, and because the alternative—redefining the criterion until a number emerges—is the failure mode this paper is about.

The first is redistribution: nickel migrating from the electrolyte-adjacent functional layer into the support, most strongly in the finest microstructure. Testing it requires an electrolyte interface to be present in both stacks of a pair. The criterion, a sustained gradient of at least 0.05 in absolute zirconia volume fraction between the two ends of a stack, is met in one pair of three, and not in the fine anode, which is the anode the hypothesis concerns: gradients are 0.0042 and 0.0240 for fine in its two states, 0.0059 and 0.0552 for medium, and 0.1035 and 0.1157 for coarse. Nickel content does vary strongly with depth

in every stack—coarse post-redox spans 0.045 to 0.494 across 0.5 μm slabs—but in fine and in pristine medium that variation is not organised as a monotone interface gradient. It is spatial heterogeneity, which is a different thing. The likeliest explanation is prosaic. These sub-volumes are 19 μm to 24 μm across the through-thickness direction against a support hundreds of micrometres thick, and there is no reason a given tomogram must straddle the interface.

The second is cumulative-strain fracture of the ceramic scaffold, motivated by the measured YSZ retention of 0.958, 0.865 and 0.000, in which the coarse anode loses ceramic percolation outright. A deterministic operator broke YSZ throats whose local redox strain, accumulated over cycles, exceeded a threshold frozen before implementation. Both declared untestable conditions fired. The coarse anode, which is the anode the hypothesis is about, yielded no extractable YSZ throat graph at all: at a 12 μm region the YSZ phase does not span the transport axis, which is itself consistent with coarse YSZ being the most fragmented phase in the dataset at a full-stack pristine P_{span} of 0.9246. And in the two anodes that did yield graphs the operator is nearly inert, breaking 1.0% to 1.2% of throats at the maximum cycle count and moving YSZ retention to 0.9995 and 0.9996 against targets of 0.958 and 0.865.

The diagnosis is a design lesson about pre-registration rather than about nickel. The strain proxy is bounded above by the frozen strain scale $\varepsilon_0 = 0.0065$, and after smoothing the per-throat values peak at 0.00413 and 0.00210. The frozen threshold of 0.010 sits above the maximum single-cycle strain by factors of 2.4 and 4.8, so only cumulative amplification lifts any throat over it. We froze an absolute threshold against a quantity whose scale we had not yet measured. The remedy is to measure the strain distribution on one region first and pre-register the threshold as a quantile of it. We did not re-pick the threshold after seeing the distribution, because doing so would be fitting a threshold to produce damage. Independent support for the strain scale itself does exist—reviews of redox cycling report approximately 1% linear expansion on the first complete oxidation of a Ni/YSZ composite [2], against the 0.65% we froze—which establishes that the strain scale was not the error even though it does not rescue the run.

4. Discussion

The six failure modes share a structure worth naming. In each case a modelling convention that is individually reasonable becomes wrong in combination with another, and the combination is invisible on the synthetic structures normally used for validation. Retaining only the largest connected component is reasonable; measuring the spanning fraction against total phase volume is reasonable; together they are circular. Requiring perfect pristine connectivity of a synthetic analogue is reasonable; it also removes the condition under which that circularity would have been visible. Validating a coarsening operator by monotone area

reduction is reasonable; so is expecting it to reproduce capillary break-up; together they are contradictory. A regular lattice is a reasonable test structure; a stochastic local operator is a reasonable model of degradation; together they cannot produce a percolation failure, because the lattice’s minimum cut is a plane and local action cannot find one.

The prescription that follows is not a general appeal to rigour but something narrower and cheaper. Validation criteria and mechanisms should be checked against each other explicitly, and preferably on structures that are imperfect in the way real ones are. Three of the six modes above would have been caught by a test structure that was merely *not* fully connected, and a fourth by asking whether the operator can in principle produce the move the mechanism requires. A synthetic object that is fully connected, periodic and smooth will pass suites that a real microstructure fails, and it will conceal precisely the interactions that matter.

4.1. Relation to continuum and discrete prior work

The impossibility of Section 3.6 is specific to voxel-swap operators validated by monotone area reduction, and it does not extend to continuum formulations. A phase-field model evolves a continuous order parameter under a free-energy functional and is not restricted to single-voxel moves, so it can pass through the transient area increase that a discrete swap cannot. This is not a hypothetical alternative for this material: phase-field simulations of Ni coarsening have been run on three-dimensional Ni–YSZ reconstructions from ptychographic nano-tomography with measured thermophysical inputs [6], validated against ex-situ imaging of annealed specimens [5], and coupled to performance models on FIB-SEM reconstructions [7], in a line going back to the three-phase simulations of Chen and co-workers [4]. That body of work is the appropriate point of comparison, and it sharpens the present claim rather than weakening it. Where a study needs coarsening, a continuum formulation is the sound choice. Where a study uses discrete voxel operators, as many connectivity and TPB studies do because they act directly on the segmented label field, the compatibility check of Eq. (2) should be run before any validity gate is adopted.

Discrete lattice methods are not thereby disqualified for this material either. A kinetic Monte Carlo model of NiO–YSZ sintering has been calibrated and validated against measured microstructure [8], and the reason it is not subject to our argument is instructive: it is a Monte Carlo method with finite-temperature acceptance, so it can cross an energy barrier that a greedy rule cannot. The barrier described here is a property of the acceptance rule, not of the lattice.

Two of our six modes have partial precedent and we do not claim them as new. The systematic error in TPB length computed from voxelised data is documented, both as a function of calculation rule [14] and of imaging resolution [15]; what we add is that a damage operator turns that static bias into a large non-monotone excursion that

no pre-damage calibration removes. That real electrodes contain isolated nickel is likewise known, and appears in the source papers for this dataset as pristine percolation values below unity [9]; what we add is the consequence, that a qualification gate demanding perfect pristine connectivity disqualifies the synthetic analogue as a place to discover metric pathologies. The remaining four—the pruning circularity, lattice minimum-cut planarity, the curvature-proxy failure and the area barrier—we have not found stated elsewhere, which is a statement about our search rather than a claim of priority.

4.2. What the negative result does and does not establish

We tested two geometry-based operator classes to a conclusion and neither reproduces the measured ordering. We have not shown that no geometric mechanism can. We have specifically not shown that agglomeration is not the physical cause of nickel percolation loss: Section 3.6 says we could not test agglomeration with this operator class under this gate, which is a statement about the method and not about the material. Nor is anything here evidence against continuum or phase-field treatments, which are subject to none of the six modes.

The limitations are the ordinary ones and they bind hard. There are three anodes, three regions of interest per anode and three damage seeds, so class comparisons are correspondingly weak, although the sign reversal in Fig. 5 is consistent across all nine regions. Pristine and degraded stacks are different specimens, so every retention magnitude carries specimen variation and no paired quantity is available. The two R_{Ni} thresholds are not independent, because the spanning cluster vanishes discontinuously and R_{Ni} crosses 0.50 and 0.10 within one round in most regions—which is itself a finding about how percolation fails in these structures. Regions are single-scale at 8 or 12 μm , the coarse anode being bounded above by a stack that is only 13.3 μm deep. Graph metrics use frozen watershed parameters. Coarse YSZ throats have a tenth-percentile inscribed diameter of roughly two voxels, below this project’s four-voxel resolution floor, so any throat-targeted ceramic fracture rests partly on sub-resolution features.

The open question is not which operator to try next. It is what class of mechanism could produce loss of long-range nickel connectivity together with retention of most of the reaction sites, which is what 80 % TPB retention at 68 % percolation retention means in the fine anode. Uniform removal thins everything at once and eliminates both together; our synthetic runs destroy 99.5 % of TPB by the time nickel percolation is lost, some 164 times more than the real anode does. Dewetting and retraction of nickel into compact particles would break long-range connectivity while comparatively preserving the nickel/zirconia contact perimeter where TPB lives. That remains the most plausible candidate, and testing it needs a formulation that can cross the barrier of Eq. (2).

5. Conclusions

1. Enforcing monotone surface-area reduction as a validity criterion structurally excludes Rayleigh-type neck break-up. On a 6-connected lattice the criterion reduces to $\text{nb}(a) \leq \text{nb}(b)$, and a sphere-neck-sphere body is already a local minimum under single-voxel swaps, so a greedy area-decreasing operator accepts no move. The criterion and the mechanism cannot both be honoured by a single-swap operator.
2. Five further failure modes were measured, each invisible on synthetic test structures: largest-component pruning makes the spanning fraction identically unity; voxel-scale erosion multiplies TPB density by 7 to 15 before destroying it; qualification gates demanding perfect pristine connectivity are unrepresentative of electrodes carrying 1.2 % to 11.2 % disconnected nickel; a jittered regular lattice cuts at exactly one full cross-section with zero variance, whereas real networks cut at 1 % to 2 % of throats; and curvature-ranked moves do not deliver the area reduction assumed of them.
3. Stochastic surface erosion applied to real microstructures with a pruning-invariant metric reverses the measured degradation ordering: the fine anode is the last to lose percolation rather than the first.
4. Neither pristine topology nor specific nickel surface area predicts the measured ordering. The coarse anode is three to five times the most topologically fragile of the three and is the best retainer in service.
5. Real nickel networks fail at 1 % to 2 % of throats through small, region-variable, non-planar cuts. A synthetic generator intended to reproduce this must reach mean coordination 3.4 to 3.9 with standard deviation 2.0 to 2.4 and a connected-pair-distance coefficient of variation near 0.41.

Data and code availability

All code, pre-registration documents, run reports and result tables are in the project repository `connectivity_margin`. Every number in this paper traces to a committed CSV: the measured signature to `out/phase6/phase6_comparison_table.csv`; the erosion bisections, minimum cuts and Spearman inputs to `out/project2/c1real_results.csv`; the metric and TPB behaviour to `c1real_o6_validity.csv` and `c1real_rni_gate.csv`; the lattice audit to `audit_ni_vulnerability.csv`; the synthetic mechanism tests to `out/platform_v2/p10_group_experiment.csv` and `out/project2/step2_02_ceiling_check.csv`, `step2_03_main.csv`, `step2_03_control.csv`; the volume-conserving gate to `o5v2_area_barrier.csv`, transcribed by `scripts/project2/o5v2_transcribe.py` from the two committed run reports; the re-distribution and ceramic-fracture branches to `h1_depth_profiles.csv`, `h1_layer_summary.csv`

and `csfm_results.csv`. Figures regenerate with `python scripts/project2/make_figures.py`.

Each pre-registration was committed before the run it governed and the chain of commits is recorded in the repository. Corrections made against our own earlier results during the work—a TPB estimator that over-counted four-fold through periodic wrap, reversed axis extents that made one audit’s cut fractions provisional until re-run, a validity check that never compared against the pristine state and printed a pass on a failing run, a pre-registered secondary-outcome rule that selected a saturating measurement point, and an internally contradictory plan specifying both Metropolis acceptance and strict area monotonicity—are logged in the run reports at the point where each was found.

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